Thiago de Paula Oliveira

Scientific Computing | Agri-Tech R&D | Technical Leadership | Reproducible Software

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PROFESSIONAL PROFILE

Scientific computing and statistics professional with 14+ years of experience spanning agricultural R&D, quantitative genetics and reproducible analytical software. Leads multidisciplinary client projects and develops reusable software, applying proof-of-concept comparison and computational benchmarking across crop, livestock and agri-tech programmes.

TECHNICAL SKILLS

- **Scientific computing and software:** R; C++ and Rcpp; SQL; Bash; Shiny; Docker; R package development; QC/ETL and reproducible analytical pipelines; HPC environments for large-scale statistical and genomic analysis.
- **Statistical modelling and machine learning:** mixed and generalised mixed models; Bayesian and multilevel modelling; generalised additive models (GAMs); support vector machines (SVM); classification and regression trees; random forests for classification and regression; state-space, longitudinal and time-series models; spatial modelling; cross-validation and predictive model evaluation; simulation; experimental design.
- **Quantitative genetics and breeding:** pedigree- and genomic-based models; quantitative genetics; genomic quality control; genotype imputation; selection indices; breeding-programme evaluation.
- **Engineering practice:** GitHub organisations and repositories; GitHub Actions; pull requests; testing; code review; profiling and benchmarking; documentation; reusable pipeline and repository conventions.
- **AI-assisted engineering:** human-in-the-loop use of AI, with systematic review and validation to improve accuracy and consistency; development of agents and reusable skills to flag potential bugs, statistical issues and reproducibility problems for human review.

PROFESSIONAL EXPERIENCE

Consultant Statistician — AbacusBio

Apr 2023–present

- Have led approximately eight client projects across crop, livestock and agri-tech programmes, co-ordinating multidisciplinary teams of three to five specialists and leading client meetings, technical discussions and analytical decision-making.
- Serve as one of AbacusBio's principal analytical software developers, creating internal R packages, reusable utilities and R/Bash/SQL pipeline templates, and using GitHub, pull-request and code-review practices to support reproducible and maintainable workflows across projects.
- Lead proof-of-concept evaluations of alternative genomic and statistical workflows, comparing predictive accuracy, cross-validation performance, concordance and runtime; translate the evidence into client recommendations that inform adoption and refinement of implementation approaches.
- Compare alternative statistical and economic modelling approaches for breeding decisions, evaluating assumptions, trade-offs and scientific implications before recommending implementation strategies aligned with client objectives.
- Provide technical review of analytical code and statistical methods before client delivery, resolving methodological risks and strengthening statistical validity, reproducibility and maintainability across analytical workflows.
- Provide technical mentoring and code-review support to approximately 15–20 multidisciplinary colleagues through workshops and collaborative development, building capability in R, Git, statistical modelling, selection-index methodology and reproducible software practices.

EARLIER EXPERIENCE

Research Fellow in Quantitative Genetics — HighlanderLab (Prof Gregor Gorjanc), The Roslin Institute, University of Edinburgh

Oct 2021–Mar 2023

- Awarded a Marie Skłodowska-Curie COFUND fellowship and developed mixed-model, Bayesian and simulation methods for sustainability-focused plant-breeding programmes.
- Developed reproducible methods to monitor genetic mean and additive genetic variance, using HPC environments for large-scale statistical and genomic analysis and working with genetics, breeding and computing teams to produce implementable analytical pipelines.

Postdoctoral Researcher in Statistics — Insight Centre for Data Analytics (Prof John Newell and Prof John Hinde), University of Galway

Apr 2019–Oct 2021

- Developed Bayesian, multilevel and forecasting models with dashboards for athlete monitoring and public-health decisions, working with clinicians and domain specialists to turn longitudinal data into interpretable monitoring outputs.

Statistical Consultant — Orreco

2019

- Developed statistical models and dashboard outputs for elite-athlete monitoring using biomarker, GPS and wellness data.

Lecturer in Statistics — University of São Paulo (ESALQ/USP)

2017–2019

- Taught experimental design, linear models, calculus and statistical programming; supervised applied projects in agriculture, genetics and reproducible computing.

SELECTED SOFTWARE

- Developed and maintained **matrixCorr** and **lcc**, CRAN R packages for method comparison and longitudinal concordance, including implementation, testing, documentation and reproducible release workflows.
- Contributed to **AlphaSimR**, **AGHmatrix**, **AlphaPart** and **Reads2Map Tools** across breeding simulation, matrix methods and genotyping workflows.

EDUCATION

- **PhD in Statistics**, University of São Paulo (ESALQ/USP), 2014–2018.
- **MSc in Statistics**, University of São Paulo (ESALQ/USP), 2012–2014.
- **BSc in Agricultural Engineering**, University of São Paulo (ESALQ/USP), 2007–2012.

SELECTED AGRICULTURAL AND GENETICS RESEARCH

- Richardson, C., Amer, P., Post, M., **Oliveira, T. d. P.**, et al. (2025). Breeding for sustainability: development of an index to reduce greenhouse gas in dairy cattle. *Animal.*
- **Oliveira, T. d. P.**, Obsteter, J., Pocrnic, I., Heslot, N. & Gorjanc, G. (2023). A method for partitioning trends in genetic mean and variance to understand breeding practices. *Genetics Selection Evolution.*
- Lara, L. A. d. C., Pocrnic, I., **Oliveira, T. d. P.**, Gaynor, C. & Gorjanc, G. (2021). Temporal and genomic analysis of additive genetic variance in breeding programmes. *Heredity.*

LANGUAGES

Portuguese (native); English (professional proficiency); Spanish (reading and listening).